

Limitation of if-else statement :

The major drawback with if condition is, it checks the condition **again and again** so it increases the burden over CPU hence we introduced switch-case statement to reduce the overhead of the CPU.

Switch Statement in java (Till JDK 17V):

It is a **selective (decision making) statement** in java so, we can select one case among the available cases.

While writing the cases, break is **optional** but if we use break then the control will move **from out of the switch body**.

We can write default so if any statement is not **matching** then default will be executed but it is optional.

In the new switch case **break and :** both are **not required**, We can replace with **->** **symbol**

If we have multiple statements then {} are compulsory.

Valid types of switch expression:

byte
short
int
char
Byte
Short
Integer
Character
enum (JDK 1.5)
String (JDK 1.7)

Invalid types are:

long
float
double
boolean

Some important points:

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- a) We cannot accept **duplicate** case value.
 - b) The label of case must be **constant OR literals**.
 - c) The break statement is used to exit from the switch block. It is optional but recommended [JDK 17 (->) break is not required]
 - d) The default case is **optional** and executes if no case matches in the switch expression.
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Old switch case :	New switch case :
<pre>case 101 : IO.println("Case 101"); break; case 102 : IO.println("Case 102"); break; default : IO.println("Invalid Case"); break; case 103 : IO.println("Case 103"); break; case 104 : IO.println("Case 104"); break;</pre>	<pre>case 101 -> IO.println("Case 101"); case 102 -> IO.println("Case 102"); case 103 -> IO.println("Case 103"); case 104 -> IO.println("Case 104"); case 105 -> IO.println("Case 105"); default -> IO.println("Invalid choice");</pre> Note : In new switch case : and break both are not reqd.

//Programs

Old switch case style :

void main()
{
 int input = Integer.parseInt(IO.readLine("Enter a number :"));
 switch(input)
 {
 case 101 :
 IO.println("Case 101");
 break;
 case 102 :
 IO.println("Case 102");
 break;
 default :
 IO.println("Invalid Case");
 break;
 case 103 :
 IO.println("Case 103");
 break;
 case 104 :
 IO.println("Case 104");
 break;
 }
}

SwitchCaseDemo1.java

/* Write a switch case program where based on the first character
select the color */

```
void main()
{
    String color = IO.readLine("Enter the color Name :");
    char ch = color.toUpperCase().charAt(0);

    switch(ch)
    {
        case 'R' -> IO.println("RED Color");
        case 'W' -> IO.println("WHITE Color");
        case 'B' -> IO.println("BLUE Color");
        case 'G' -> IO.println("GREEN Color");
        default -> IO.println("Invalid color choice");
    }

    IO.println("Out of switch body");
}
```

SwitchDemo2.java

void main()
{

 IO.println("\t\t**Main Menu**\n");
 IO.println("\t\t**100 Police**\n");
 IO.println("\t\t**101 Fire**\n");
 IO.println("\t\t**102 Ambulance**\n");
 IO.println("\t\t**139 Railway**\n");
 IO.println("\t\t**181 Women's Helpline**\n");

 IO.print("Enter your choice :");
 int choice = Integer.parseInt(IO.readLine());

 switch(choice)
 {
 case 100 -> IO.println("Police Services");

 case 101 -> IO.println("Fire Services");

 case 102 -> IO.println("Ambulance Services");

 case 139 -> IO.println("Railway Enquiry");

 case 181 -> IO.println("Women's Helpline");

 default -> IO.println("Your choice is wrong");
 }
}

SwitchDemo3.java

//Passing String in the switch case [Allowed from JDK 1.7]

```
void main()
{
    IO.print("Enter the season name :");
    String season = IO.readLine().toLowerCase();

    switch(season) //1.7V
    {
        case "summer" -> IO.println("Summer season");
        case "winter" -> IO.println("Winter season");
        case "spring" -> IO.println("Spring season");
        case "rainy" -> IO.println("Rainy season");
        default -> IO.println("No valid season");
    }
}
```

Switch Case with Grouping in Java:

In switch statement, It is possible to combine **multiple case labels** to execute one common block of code. It must be separated by , (comma)

We should use when **multiple values (cases)** should result in the same action.

SwitchDemo4.java

void main()
{
 IO.print("Enter an animal Name :");
 String animal = IO.readLine().toLowerCase();

 switch(animal)
 {
 case "cat", "dog", "cow" ->
 {
 IO.println("Domestic Animal");
 }

 case "lion", "tiger", "fox" ->
 {
 IO.println("Wild Animal");
 }
 }
}